Zackary Rackauckas

Speech & Language Researcher — Pronunciation Assessment, Expressive TTS, Conversational Agents

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Education

Columbia Engineering *New York, NY*

MS Computer Science, Thesis Track *December 2025*

- Thesis: "Stylized Conversational Agents for Japanese Language Learning: System Design, User Engagement, and Expressive TTS Evaluation." Advisor: Prof. Julia Hirschberg.
- GPA: 4.0/4.0

Swarthmore College *Swarthmore, PA*

BA Mathematics *May 2023*

- Minors in Computer Science and Japanese.
- GPA: 3.8/4.0.

Research Experience

Harvard Graduate School of Education *Cambridge, MA*

Predoctoral Research Assistant · Supervisor: Dr. Ying Xu *Jun. 2026 – Present*

- Co-developing PEDBench, an open-source benchmark for evaluating proprietary and open-source audio-to-phoneme models on public, authentic, and synthetic children's speech, characterizing failure modes and the extent to which current models preserve the phonetic errors relevant to pronunciation assessment.
- Investigating audio-language models for mispronunciation detection by fine-tuning Qwen2.5-Omni-7B and Audio-Flamingo 3 with LoRA across TIMIT, L2-ARCTIC, and synthetic speech.
- Developing phoneme-controlled text-to-speech models that synthesize natural speech directly from CMU-39 phoneme sequences, enabling controlled generation of specified pronunciations.
- Designing a child-facing pronunciation-tutoring system that delivers sentence-, word-, and phoneme-level feedback from ASR and mispronunciation detection to learners and teachers.

University of California, Irvine – School of Education *Irvine, CA*

Research Associate · Supervisor: Dr. Mark Warschauer *May 2026 – Present*

- Evaluating multimodal language models on a lab-collected corpus of children's speech to characterize transcription performance on child speakers; presented findings in the research talk "ASR for Children" (Digital Learning Lab, 2026).
- Investigating multimodal language models within Converse to Learn, a conversational-agent pipeline for early science education.

Columbia University – Dept. of Computer Science *New York, NY*

Research Assistant & MS Thesis Student · Advisor: Dr. Julia Hirschberg *Aug. 2024 – Dec. 2025*

- Developed AdaptLingo, a speech-to-speech English practice system that combines prosody-based fluency estimation with constrained LLM decoding to adapt a conversational agent's responses to learner proficiency (Interspeech 2026 Show & Tell; JALTCALL 2025; Columbia Data Science Day).
- Developed a method for detecting and correcting learner pronunciation errors using flow-matching text-to-speech and speech editing.
- Evaluated VITS and Style-BERT-VITS2 as candidate architectures for expressive Japanese character text-to-speech in stylized conversational agents (IEEE UEMCON 2025).
- Proposed VoxRAG, a transcription-free retrieval-augmented generation framework for spoken question answering that retrieves directly from speech rather than relying on an intermediate transcript (MAGMaR Workshop at ACL 2025).
- Co-authored a survey of automated presentation-coaching systems synthesizing existing methods, system designs, and open research challenges (BEA Workshop at ACL 2026).

University of Tokyo – Dept. of Electrical Engineering and Information Systems *Tokyo, Japan*

Visiting Research Intern · Supervisor: Dr. Nobuaki Minematsu *May 2025 – Jul. 2025*

- Developed Re:Member, a multimodal dialogue system that generates emotionally grounded questions from videos of learners' personal memories for Japanese conversation practice (HCI+NLP Workshop at EMNLP 2025).
- Mentored a BS student's research on LangInLab, vision- and voice-enabled agents for language support in engineering laboratory instruction ahead of UTokyo's transition to English-medium instruction (HAI 2025).

RoleGaku (Jouzu) *Irvine, CA*

Founder & Independent Researcher *Dec. 2023 – Present*

- Developed Jouzu as an experimental platform for studying spoken interaction with stylized conversational agents in Japanese language learning, combining voiced LLM characters, expressive Japanese TTS, pronunciation feedback, and interactive vocabulary support, and released it as a mobile application to real learners.
- Designed and conducted in-the-wild studies of how anime-inspired personas, character voices, and culturally grounded interface design influence learner engagement and language practice (JALTCALL Trends 2025; arXiv preprint with J. Hirschberg; Interspeech 2026 Show & Tell; JALTCALL 2024).
- Led an interdisciplinary team spanning engineering, art, writing, and voice acting.

Infineon Technologies *San Jose, CA*

Intern *Aug. 2023 – Apr. 2024*

- Proposed RAG-Fusion, a multi-query retrieval-augmented generation method (IJNLC 2024), and evaluated it with RAGElo, an automated Elo-based framework for comparing RAG systems on answer accuracy, retrieval reliability, and hallucination across heterogeneous query tasks (LLM4Eval Workshop at SIGIR 2024).
- Developed a document-grounded RAG assistant for product documentation that cut engineering search time from minutes to seconds, and compared RAG configurations across enterprise use cases.

Chapman University – Dept. of Computer Science *Orange, CA*

Machine Learning Research Associate · Supervisor: Dr. Erik Linstead *Jun. 2022 – Aug. 2022*

- Measured shifts in public opinion on Twitter before and after the overturning of *Roe v. Wade* using sentiment analysis.

Swarthmore College – Dept. of Modern Languages and Literature *Swarthmore, PA*

Research Assistant · Supervisor: Dr. John Bundschuh *Jun. 2022 – Aug. 2022*

- Digitized archaic Japanese Buddhist sutras to support computational study of diachronic change in Japanese, with a focus on historical variation in verb morphology.

Stanford University School of Medicine *Stanford, CA*

Intern *Jul. 2021 – Dec. 2021*

- Translated evidence-based suicide-reporting guidelines (the Tool for Evaluation of Media Portrayals of Suicide) into an interactive assessment tool that evaluates draft news articles and gives journalists targeted revision guidance.

Publications

Conference and Workshop Papers (Peer-Reviewed)

[C1] Wen Liang, Li Siyan, **Zackary Rackauckas**, Julia Hirschberg. 2026. A Survey of Automated Presentation Coaching: Systems, Methods, and Open Challenges. In *Proceedings of the 21st Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2026) at the Annual Meeting of the Association for Computational Linguistics **(ACL 2026)***, 34–47. URL: `https://aclanthology.org/2026.bea-1.4/`

[C2] **Zackary Rackauckas**, Nobuaki Minematsu, Julia Hirschberg. 2025. Re:Member: Emotional Question Generation from Personal Memories. In *Proceedings of the 4th Workshop on Bridging Human–Computer Interaction and Natural Language Processing (HCI+NLP) at the Conference on Empirical Methods in Natural Language Processing **(EMNLP 2025)** (Oral)*. URL: `https://aclanthology.org/2025.hcinlp-1.13/`

[C3] **Zackary Rackauckas**, Julia Hirschberg. 2025. Comparative Evaluation of Expressive Japanese Character Text-to-Speech with VITS and Style-BERT-VITS2. In *Proceedings of the 2025 IEEE 16th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference **(IEEE UEMCON 2025)*** (Oral). URL: `https://ieeexplore.ieee.org/document/11267532`. **Best Presentation Award.**

[C4] **Zackary Rackauckas**, Julia Hirschberg. 2025. VoxRAG: A Step Toward Transcription-Free RAG Systems in Spoken Question Answering. In *Proceedings of the 1st Workshop on Multimodal Augmented Generation via Multimodal Retrieval (MAGMaR 2025) at the Annual Meeting of the Association for Computational Linguistics **(ACL 2025)***. URL: `https://aclanthology.org/2025.magmar-1.3/`

[C5] **Zackary Rackauckas**, Arthur Câmara, Jakub Zavrel. 2024. Evaluating RAG-Fusion with RAGElo: An Automated Elo-Based Framework. In *Proceedings of the 1st Workshop on Large Language Models for Evaluation in Information Retrieval (LLM4Eval) at the ACM SIGIR Conference on Research and Development in Information Retrieval **(SIGIR 2024)***. URL: https://ceur-ws.org/Vol-3752/paper6.pdf

Demonstrations and Extended Abstracts (Peer-Reviewed)

[D1] **Zackary Rackauckas**, Li Siyan, Julia Hirschberg. 2026. AdaptLingo: A Speech-to-Speech English Practice System with Fluency-Adaptive Responses. In *Proc. Interspeech 2026*, 2382–2383. URL: https://www.isca-archive.org/interspeech_2026/rackauckas26_interspeech.pdf

[D2] **Zackary Rackauckas**. 2026. A Speech-First Character Interface for Stylized Japanese Dialogue Practice. In *Proc. Interspeech 2026*, 2388–2389. URL: https://www.isca-archive.org/interspeech_2026/rackauckas26b_interspeech.pdf

[D3] Masanori Shigi, **Zackary Rackauckas**, Yuka Akiyama, Nobuaki Minematsu. 2025. LangInLab: Augmenting Engineering Lab Instruction with Vision- and Voice-Enabled AI Agents for Language Learning. In *Proceedings of the 13th International Conference on Human-Agent Interaction **(HAI 2025)***. URL: https://doi.org/10.1145/3765766.3765844

Journal Articles (Peer-Reviewed)

[J1] **Zackary Rackauckas**. 2025. Learner Interactions with Jouzu, the Mobile Application with Conversational Characters. *JALTCALL Trends*, 1(1). URL: https://doi.org/10.29140/jct.v1n1.2164

[J2] **Zackary Rackauckas**. 2024. RAG-Fusion: A New Take on Retrieval-Augmented Generation. *International Journal on Natural Language Computing*, 13(1), 37–47. URL: https://doi.org/10.5121/ijnlc.2024.13103.

Preprints and Manuscripts Under Review

[P2] **Zackary Rackauckas**, Julia Hirschberg. 2025. Learning Japanese with Jouzu: Interaction Outcomes with Stylized Dialogue Fictional Agents. URL: https://arxiv.org/abs/2507.06483

Invited Talks and Presentations

Zackary Rackauckas. 2026. "Speech Technology for Language Learning From Adaptive Interaction to Phonetic Error Detection." Research talk: *Johns Hopkins University SMILE Lab*, Baltimore, USA.

Zackary Rackauckas. 2026. "Speech Technology for Language Learning From Adaptive Interaction to Phonetic Error Detection." Research talk: *UCLA Speech Processing and Audio Perception Laboratory*, Los Angeles, USA.

Zackary Rackauckas. 2026. "ASR for Children." Research talk: *Digital Learning Lab*, Irvine, USA.

Zackary Rackauckas. 2025. "Demonstrating an AI Chatbot that Adapts to Users' English Proficiency with End-to-End Speech." Workshop presentation: *Japan Association for Language Teaching–Computer-Assisted Language Learning (JALTCALL 2025)*, Tokyo, Japan.

Zackary Rackauckas. 2025. "Information Retrieval and Dialogue Systems for Computer-Assisted Language Learning." Research talk: *GAVO Minematsu & Saito Laboratory*, Tokyo, Japan.

Zackary Rackauckas, Riya Raj, Siyan Li, Julia Hirschberg. 2025. "AdaptLingo: An Adaptive Speech-to-Speech Conversational Agent Tailored to Users' English Proficiency Levels." Poster presentation: *Columbia Data Science Day*, New York, USA.

Zackary Rackauckas, Riya Raj, Siyan Li, Julia Hirschberg. 2025. "AdaptLingo: An Adaptive Speech-to-Speech Conversational Agent Tailored to Users' English Proficiency Levels." Poster presentation: *Columbia AI Summit*, New York, USA.

Zackary Rackauckas. 2024. "Jouzu: Voiced Character Language Chatbots for Japanese Learning." Research talk: *Computer-Enabled Abilities Laboratory*, New York, USA.

Zackary Rackauckas. 2024. "Voiced Character Language Chatbots for Japanese Learning." Oral presentation: *Japan Association for Language Teaching–Computer-Assisted Language Learning (JALTCALL 2024)*, Nagoya, Japan.

Awards

2025 **Best Presentation Award**, IEEE UEMCON

2025 **Friends of Global UTokyo Leadership Award**, The University of Tokyo *$4,000*

2021–2023 **Academic Honor Roll, Varsity Golf**, Centennial Conference (NCAA Division III)

Professional Service & Memberships

Reviewing and Program Committees

2026 **Concurrency and Computation: Practice and Experience**, Reviewer

2025 **International Conference on Human-Agent Interaction (HAI 2025)**, Reviewer

2025 **Workshop on LLMs for Evaluation in Information Retrieval (LLM4Eval), 2nd and 3rd editions**, Reviewer

2025 **JALTCALL Trends**, Reviewer, inaugural issue

Memberships

2025– **Association for Computational Linguistics (ACL)**, Student Member

2024– **IEEE, IEEE Computer Society, and IEEE Young Professionals**, Member

Skills & Languages

Speech and Language ML

- PyTorch, Hugging Face Transformers and PEFT (LoRA fine-tuning of audio-language models: Qwen2.5-Omni, Audio-Flamingo 3), ASR and phoneme recognition, TTS (VITS, Style-BERT-VITS2, flow-matching models), speech editing, prosody analysis, constrained LLM decoding, retrieval-augmented generation, scikit-learn, NLTK, Praat.

Research Methods

- Benchmark design for speech models, in-the-wild user studies of conversational agents, human evaluation of expressive TTS and dialogue systems, automated LLM-based evaluation (Elo-style ranking), synthetic speech data generation.

Programming

- Python, JavaScript, C/C++, SQL, R.

Systems and Tools

- React, React Native, WebRTC, Flask, PostgreSQL, REST APIs, cloud deployment, Git, LaTeX.

Languages

- English (native); Japanese (advanced, JLPT N2, 2024); Spanish (near-native); Mandarin Chinese (intermediate).

Mentoring

2025 **Masanori Shigi**, BS Student, The University of Tokyo

2024–2025 **Riya Raj**, MS Student, Columbia University